BCH212 Medical Biochemistry

Simulation Examination II

Lectures 10 – 21 | 160 Questions with Detailed Explanations

Questions marked ★ are from past papers — prioritise these!

Topic distribution: Lec10 (8Q) | Lec11 (8Q) | Lec12 (10Q) | Lec13 (8Q) | Lec14 (8Q) | Lec15 (12Q) | Lec16 (8Q) | Lec17 (8Q) | Lec18 (12Q) | Lec19 (10Q) | Lec20 (6Q) | Lec21 (18Q)

LECTURE 10: DNA & RNA SYNTHESIS

Q1. ★ [PAST PAPER]   DNA replication results in two daughter DNA molecules. Which of the following correctly describes the composition of the new strands?

A. Each molecule has 2 new strands
B. Each molecule has 1 new and 1 original strand
C. Both molecules contain two original strands
D. Both molecules have two new strands
E. None of the above

✔ Answer: B
DNA replication is semi-conservative. Each daughter molecule retains one parental (original) strand and gains one newly synthesised strand. Meselson-Stahl experiment confirmed this. (★ Past paper: SA2 Med34 Q4, BCH212 Re-exam Q7)

Q2. ★ [PAST PAPER]   Although nucleotide addition during DNA replication in prokaryotes is ~20x faster than in eukaryotes, why can larger eukaryotic genomes be replicated in a time-effective manner?

A. Eukaryotes have less genetic material to transcribe
B. Eukaryotes have fewer polymerase types
C. Eukaryotes have helicase which can more easily unwind DNA strands
D. Eukaryotes have multiple origins of replication
E. Eukaryotes do not have exons

✔ Answer: D
Eukaryotes initiate replication simultaneously at thousands of origins (ori) along each chromosome — this compensates for the slower polymerisation rate per site. Prokaryotes have only one origin per chromosome. (★ Past paper: SA2 Med34 Q3)

Q3. ★ [PAST PAPER]   Okazaki fragments are best defined as:

A. Discontinuous sequences of DNA nucleotides synthesised on the lagging strand
B. Long continuous segments of the leading strand
C. Short RNA primers on the leading strand
D. Protein-DNA complexes at the replication fork
E. Telomeric repeat sequences

✔ Answer: A
Because DNA polymerase can only synthesise 5'→3', the lagging strand is copied in short, discontinuous segments called Okazaki fragments, each starting with an RNA primer. These are later joined by DNA ligase. (★ Past paper: SA2 Med34 Q5)

Q4. ★ [PAST PAPER] Which of the following is true about telomeres?

A. Telomeres are at the centre of eukaryotic chromosomes
B. Telomeres are degraded by telomerase
C. Telomeres do not contain functional genes
D. Telomeres become longer with every round of replication
E. All of the above are false

✔ Answer: C

Telomeres are repetitive sequences (TTAGGG in humans) at chromosome ends. They protect chromosomes from degradation. They do NOT contain coding genes. Telomerase elongates them (not degrades). Without telomerase, telomeres shorten with each division. (★ Past paper: SA2 Med34 Q1)

Q5. ★ [PAST PAPER] Which type of DNA repair is used when a single nucleotide is damaged by alkylation?

A. Nucleotide excision repair (NER)
B. Base excision repair (BER)
C. Mismatch repair (MMR)
D. RNA splicing
E. Homologous recombination

✔ Answer: B

BER removes a single damaged base (e.g., alkylated, oxidised, or deaminated bases) and replaces it. NER removes bulky lesions (e.g., thymine dimers from UV). MMR corrects mispaired bases from replication errors. (★ Past paper: Key Simulation Q1)

Q6. ★ [PAST PAPER] Which drug used in cancer chemotherapy inhibits mammalian topoisomerase?

A. Rifampicin
B. Amanitin
C. Tetracycline
D. Adriamycin (Doxorubicin)
E. Chloramphenicol

✔ Answer: D

Adriamycin (Doxorubicin) inhibits topoisomerase II, preventing DNA strand re-ligation, causing double-strand breaks in rapidly dividing cancer cells. Rifampicin inhibits bacterial RNA polymerase; Amanitin inhibits eukaryotic RNA-pol II. (★ Past paper: Key Simulation Q3)

Q7. ★ [PAST PAPER] Xeroderma pigmentosum results from deficiency of which repair mechanism?

A. Base excision repair
B. Mismatch repair
C. Nucleotide excision repair
D. Homologous recombination
E. Non-homologous end joining

✔ Answer: C

XP is caused by mutations in NER genes. NER removes bulky lesions such as thymine dimers (TT dimers) caused by UV light — these distort the DNA helix. Without NER, UV damage accumulates causing skin cancers. (★ Past paper: FA2 Sec11 Q3)

Q8. Which of the following correctly distinguishes replication from transcription?

A. Transcription requires a primer; replication does not
B. Replication requires a primer; transcription does not
C. Both replication and transcription use the same enzyme
D. The substrate in transcription is dNTP
E. Both processes use the entire DNA molecule as template

✔ Answer: B

DNA polymerase cannot start a new strand from scratch — it requires a short RNA primer synthesised by primase. RNA polymerase can initiate transcription de novo (no primer needed). Transcription uses NTPs (ribonucleotides), not dNTPs.

LECTURE 11: PROTEIN SYNTHESIS & TRANSPORT

Q9. ★ [PAST PAPER] Which release factor recognises the UGA stop codon during translation in bacteria?

A. RF1
B. RF2
C. RF3
D. EF-G
E. EF-Tu

✔ Answer: B

In bacteria: RF1 recognises UAA and UAG; RF2 recognises UAA and UGA. RF3 is a GTPase that helps recycle RF1/RF2. In eukaryotes, a single release factor (eRF1) recognises all three stop codons. (★ Past paper: SA2 Med34 Q2, Re-exam Q18)

Q10. ★ [PAST PAPER] Which biomolecule is responsible for synthesising aminoacyl-tRNA?

A. 16S rRNA
B. 23S rRNA
C. 28S rRNA
D. Aminoacyl-tRNA synthase (incorrect term)
E. Aminoacyl-tRNA synthetase

✔ Answer: E

Aminoacyl-tRNA synthetases (one for each amino acid) catalyse the attachment of the correct amino acid to its specific tRNA using ATP. The enzyme charges the tRNA in two steps: (1) amino acid + ATP → aminoacyl-AMP; (2) aminoacyl-AMP + tRNA → aminoacyl-tRNA. (★ Past paper: FA2 Sec11 Q5)

Q11. ★ [PAST PAPER] A patient ate castor beans and developed protein synthesis failure. The toxin in castor beans (ricin) acts by:

A. Inhibiting aminoacyl-tRNA binding to the ribosome A site
B. Cleaving the large ribosomal subunit RNA (28S rRNA)
C. Inhibiting peptidyl transferase activity
D. Blocking translocation by EF-2
E. Inhibiting initiation factor eIF2

✔ Answer: B

Ricin is an N-glycosidase that depurinates a specific adenine in 28S rRNA of the large ribosomal subunit, permanently inactivating it. This halts translation at the elongation stage. (★ Past paper: SA2 Med34 Q1, Re-exam Q17)

Q12. ★ [PAST PAPER] What is the first aminoacyl-tRNA binding site during initiation of eukaryotic translation?

A. E site
B. A site
C. P site
D. Anywhere on the ribosome
E. Exit tunnel

✔ Answer: C

The initiator Met-tRNA directly occupies the P (peptidyl) site during initiation — all subsequent aminoacyl-tRNAs enter at the A (aminoacyl) site. In prokaryotes, the initiator is fMet-tRNA, which also binds P site. (★ Past paper: SA2 Med34 Q8, Re-exam Q21)

Q13. ★ [PAST PAPER] Which antibiotic inhibits translation in BOTH prokaryotes and eukaryotes?

A. Chloramphenicol
B. Tetracycline
C. Puromycin
D. Streptomycin
E. Erythromycin

✔ Answer: C

Puromycin mimics aminoacyl-tRNA and incorporates into the growing peptide chain, causing premature termination. Unlike most antibiotics, it affects both prokaryotic (70S) and eukaryotic (80S) ribosomes. (★ Past paper: Key Simulation Q6)

Q14. ★ [PAST PAPER] Which molecule guides translated proteins destined for modification to the endoplasmic reticulum?

A. RF1
B. RF2
C. Signal Recognition Particle (SRP)
D. EF-2
E. IF2

✔ Answer: C

Proteins destined for secretion, membranes, or organelles carry an N-terminal signal peptide. SRP recognises this peptide co-translationally and escorts the ribosome-mRNA-nascent protein complex to the ER membrane. (★ Past paper: SA2 MED35 Q18)

Q15. Which of the following proteins does NOT require a signal sequence for proper targeting?

A. Insulin
B. Cytochrome P450
C. GLUT4
D. Growth hormone
E. Pyruvate kinase

✔ Answer: E

Pyruvate kinase is a cytosolic enzyme — it stays in the cytoplasm and does not need a signal sequence. Insulin, growth hormone, and secreted proteins require signal sequences. GLUT4 is a membrane transporter also requiring targeting. (★ Past paper: SA2 Med34 Q7)

Q16. Tetracycline exerts its antibacterial effect by:

A. Inhibiting peptidyl transferase in the 50S subunit
B. Blocking aminoacyl-tRNA entry into the A site of the 30S subunit
C. Causing misreading of the mRNA template
D. Inhibiting translocation by EF-G
E. Cleaving 16S rRNA

✔ Answer: B

Tetracycline binds the 30S ribosomal subunit and blocks the A site, preventing aminoacyl-tRNA from binding. This halts elongation. Chloramphenicol inhibits peptidyl transferase (50S). Streptomycin causes misreading.

LECTURE 12: HUMAN GENOME & GENE REGULATION

Q17. ★ [PAST PAPER] Fragile X syndrome is caused by expansion of CGG repeats in the FMR1 gene. Which process is primarily disrupted leading to disease?

A. Splicing
B. Translation
C. Replication
D. Transcription
E. Post-translational modification

✔ Answer: D

When CGG repeats exceed ~200, the FMR1 promoter and repeat region become hypermethylated. This silences the gene — transcription is blocked. FMRP protein (needed for synaptic development) is absent, causing intellectual disability. (★ Past paper: FA2 Med33-34 Q7, Re-exam Q1)

Q18. ★ [PAST PAPER] A 25-year-old woman presents with infertility, no menarche, webbed neck, and broad chest. What is her most likely karyotype?

A. 46, XO
B. 45, XO
C. 47, XXX
D. 47, XXY
E. 46, XX, t(14;21)

✔ Answer: B

These features describe Turner syndrome: short stature, webbed neck, broad chest, infertility, no menstrual periods, and normal intelligence. Turner syndrome karyotype is 45,XO (monosomy X). Note: 46,XO does not exist — 46 with XO is written 46,XX,del(Xp). (★ Past paper: FA2 Sec11 Q9, Key Simulation Q1, Re-exam Q2)

Q19. ★ [PAST PAPER] Which protein complex bridges activators bound to distal enhancers with transcription factors near the core promoter in eukaryotes?

A. Housekeeping activator
B. Sigma factor
C. Mediator complex
D. Transcription factor IID (TFIID)
E. Spliceosome

✔ Answer: C

The Mediator is a large multi-subunit complex that acts as a 'bridge' — it physically connects activator proteins bound to enhancer sequences (far from the gene) with general transcription factors assembled at the core promoter (TATA box region), facilitating RNA polymerase II recruitment. (★ Past paper: Key Simulation Q5)

Q20. ★ [PAST PAPER]   A patient with promoter mutation leading to decreased TNF gene expression. Which of the following is most directly altered?

A. DNA polymerase activity
B. RNA polymerase binding efficiency
C. mRNA production rate
D. Protein production
E. Promoter site binding affinity

✔ Answer: E

The promoter is the regulatory DNA sequence where transcription factors assemble and RNA polymerase binds. A promoter mutation directly alters the binding affinity of these proteins to the promoter — this is the FIRST event. mRNA and protein production are downstream consequences. (★ Past paper: SA2 Med34, BCH212 Re-exam Q11)

Q21. ★ [PAST PAPER]   In the lac operon, for MAXIMAL expression of structural genes, which condition must be met?

A. High cAMP and low lactose
B. Low glucose and lactose present (CAP site occupied)
C. High glucose and high lactose
D. The attenuation stem-loop can form
E. Repressor protein is bound to operator

✔ Answer: B

Maximal lac operon expression requires two conditions simultaneously: (1) lactose present — allolactose binds and inactivates the repressor (operator is free); (2) glucose absent — low glucose → high cAMP → cAMP-CAP complex binds CAP site → enhances RNA polymerase binding. (★ Past paper: Key Simulation Q6)

Q22. ★ [PAST PAPER]   Klinefelter syndrome (47, XXY) results from which mechanism?

A. Deletion of X chromosome
B. Inversion of chromosome X
C. Non-disjunction during meiosis
D. Trinucleotide repeat expansion
E. Robertsonian translocation

✔ Answer: C

Klinefelter syndrome arises when chromosomes fail to separate properly during meiosis I or II (non-disjunction), resulting in a sperm or egg with an extra sex chromosome. Features include tall stature, small testes, gynecomastia, infertility, and usually normal intelligence. (★ Past paper: Lec15, SA2 Med36)

Q23.   A riboswitch is found in the 5' UTR of a bacterial mRNA. When its target metabolite (TPP) binds, a termination stem-loop forms. This riboswitch regulates gene expression at which level?

A. Post-translation
B. Post-transcription
C. Translation
D. Transcription
E. Replication

✔ Answer: D

When TPP binds the aptamer domain, the expression platform forms a Rho-independent terminator stem-loop followed by a poly-U track, causing RNA polymerase to fall off — transcription is terminated prematurely. This is transcriptional regulation. If it blocked the RBS instead, it would be translational regulation. (★ Past paper: SA2 Med34 Q5 on riboswitch)

Q24. ★ [PAST PAPER] Estrogen binds to the estrogen receptor (ER), which then binds to proximal promoters. What type of protein is the ER?

A. Nuclear receptor
B. Cell membrane receptor
C. Membrane enzyme
D. Adhesion molecule
E. Transporter

✔ Answer: A

Estrogen is a lipid-soluble steroid hormone that crosses the cell membrane and binds to intracellular receptors. The estrogen receptor is a nuclear receptor — upon ligand binding it dimerises, translocates to the nucleus, and acts as a transcription factor binding to oestrogen response elements (EREs). (★ Past paper: SA2 MED35 Q12)

Q25. Which of the following describes an AGO2-mediated mechanism of gene silencing?

A. It methylates histone H3K27
B. It degrades target mRNA via perfect complementarity (siRNA pathway)
C. It blocks transcription of the target gene
D. It exports pre-miRNA from the nucleus
E. It converts pri-miRNA to pre-miRNA

✔ Answer: B

AGO2 is the only human Argonaute protein with slicer (endonuclease) activity. In the siRNA pathway, perfect complementarity between siRNA and mRNA guides AGO2 to cleave the target mRNA. AGO1 mediates translational repression with imperfect complementarity (miRNA). Drosha converts pri-miRNA to pre-miRNA; Exportin transports pre-miRNA; Dicer processes pre-miRNA.

Q26. Which technique would be MOST appropriate to detect the chromosomal location of a specific gene on chromosome 21 in a patient with Down syndrome?

A. Southern blot
B. Northern blot
C. Western blot
D. Fluorescence In Situ Hybridisation (FISH)
E. PCR-RFLP

✔ Answer: D

FISH uses fluorescent probes complementary to a specific chromosomal region. The probes hybridise to denatured chromosomes on a slide and can be visualised under fluorescence microscopy. FISH can detect trisomy 21 (3 signals instead of 2) and gene location simultaneously.

LECTURE 13: THALASSEMIA

Q27. ★ [PAST PAPER] What is the genotype of Hb Bart's hydrops fetalis?

A. --/-α (alpha-thal1/alpha-thal2)
B. -α/-α (homozygous alpha-thal2)
C. --/-- (homozygous alpha-thal1)
D. αα/-- (heterozygous alpha-thal1)
E. α^CS α/--

✔ Answer: C

Hb Bart's hydrops fetalis is the most severe form of alpha-thalassemia — ALL FOUR alpha-globin genes are absent/non-functional (--/--). Without any alpha chains, gamma chains form tetramers (gamma-4 = Hb Bart's). This has extremely high O2 affinity and causes fatal intrauterine hypoxia. (★ Past paper: SA2 Med34, SA2 MED35, Med36, Re-exam)

Q28. ★ [PAST PAPER] Which genotype results in Hb H disease?

A. --/--
B. --/-α (alpha-thal1/alpha-thal2)
C. -α/-α
D. αα/--
E. αα/αα

✔ Answer: B

Hb H disease = 3 of 4 alpha genes deleted (--/-α). Only 1 functional alpha gene remains. Excess beta chains form beta-4 tetramers = Hb H. Causes moderate-to-severe anaemia. Note: --/α^CS α (HbH Constant Spring) is a severe variant. (★ Past paper: SA2 Med34, Key Simulation Q4)

Q29. ★ [PAST PAPER] Which substance pharmacologically stimulates HbF production in beta-thalassemia?

A. Deferoxamine
B. Deferiprone
C. Folate
D. Hydroxyurea
E. Warfarin

✔ Answer: D

Hydroxyurea (and also 5-azacytidine, butyrate compounds, erythropoietin) pharmacologically stimulate HbF (alpha-2 gamma-2) production. Increased HbF compensates for deficient adult HbA. Deferoxamine and deferiprone are iron chelators used to treat iron overload from transfusions. (★ Past paper: SA2 Med34, Key Simulation Q2&Q5;)

Q30. ★ [PAST PAPER] Father has beta-thalassemia trait (carrier), mother has homozygous HbE (beta-E/beta-E). What is the percentage probability that their child will have thalassemia disease?

A. 0%
B. 25%
C. 50%
D. 75%
E. 100%

✔ Answer: C

Father is beta/beta0 (carrier); mother is betaE/betaE. Each child gets one gene from each parent. From father: 50% chance of getting beta0 allele. From mother: 100% chance of getting betaE. Beta0/betaE = thalassemia disease (beta-thal/HbE). The other 50% get beta/betaE = HbE trait (mild). Therefore 50% have thalassemia disease. (★ Past paper: SA2 Med34 Q2, Med36 Q5)

Q31. ★ [PAST PAPER] Hemoglobin Constant Spring is caused by mutation in which codon?

A. Codon 26, GAG → AAG
B. Codon 41/42, deletion TTCT
C. Codon 142, UAA → CAA
D. Codon 6, GAG → GTG
E. Codon 17, AAG → UAG

✔ Answer: C

HbCS: the normal stop codon (UAA) at position 142 of alpha-globin mutates to CAA (Glutamine). Translation continues past the normal stop, producing an abnormally LONG alpha-chain (172 aa instead of 141). This is a non-deletion alpha-thalassemia mutation. (★ Past paper: SA2 MED35, Key Simulation Q4)

Q32. ★ [PAST PAPER] Which hemoglobin is found predominantly in FETAL blood?

A. alpha-2 beta-2 (HbA)
B. alpha-2 gamma-2 (HbF)
C. alpha-2 delta-2 (HbA2)
D. zeta-2 epsilon-2 (Hb Gower 1)
E. zeta-2 gamma-2 (Hb Portland)

✔ Answer: B

HbF (alpha-2 gamma-2) is the predominant haemoglobin from 2nd trimester through birth. It has higher O2 affinity than HbA, facilitating O2 transfer from maternal to fetal blood. At birth: ~97% HbF, rapidly replaced by HbA by 6-12 months. (★ Past paper: SA2 Med34, Med36, SA2 MED35 Q50/Q80)

Q33. ★ [PAST PAPER] Repeated blood transfusions in thalassemia patients commonly lead to which serious complication?

A. Hyperkalemia
B. Iron overload (hemosiderosis)
C. Folate deficiency
D. Elevated HbF
E. Thrombocytopenia

✔ Answer: B

Each unit of transfused blood contains ~200-250 mg iron. The body has no mechanism to excrete excess iron. Iron accumulates in heart, liver, pancreas, and endocrine glands causing organ damage (cardiomyopathy, cirrhosis, diabetes). Iron chelators (deferoxamine, deferiprone, deferasirox) are used to manage iron overload. (★ Past paper: Med36, Lec13 notes)

Q34. Which hemoglobin is found in the EMBRYONIC phase (earliest development)?

A. alpha-2 beta-2
B. alpha-2 gamma-2
C. alpha-2 delta-2
D. alpha-2 zeta-2
E. zeta-2 epsilon-2

✔ Answer: E

In the embryonic yolk sac phase (first 8 weeks): Hb Gower 1 (zeta-2 epsilon-2) and Hb Gower 2 (alpha-2 epsilon-2) and Hb Portland (zeta-2 gamma-2) are produced. Zeta (zeta) is the embryonic alpha-like chain; epsilon (epsilon) is the embryonic beta-like chain. (★ Past paper: SA2 Med34 Q1, Key Simulation Q6, BCH212 Re-exam Q25)

LECTURE 14: APOPTOSIS

Q35. ★ [PAST PAPER] Which of the following is a characteristic feature of APOPTOSIS but NOT necrosis?

A. Cell swelling
B. Nuclear fragmentation and chromatin condensation
C. Membrane rupture with release of cellular contents
D. Significant inflammatory response
E. Random DNA degradation

✔ Answer: B

Apoptosis (programmed death): cell SHRINKAGE, chromatin condensation, nuclear fragmentation, membrane blebbing, apoptotic body formation, phagocytosed without inflammation. Necrosis (passive death): cell SWELLING, membrane rupture, release of contents, triggers inflammation. (★ Past paper: SA2 Med34 Q1, Med36 Q4, Re-exam Q6)

Q36. ★ [PAST PAPER] Which of the following components is NOT part of the apoptosome?

A. Cytochrome C
B. APAF-1
C. dATP
D. Pro-caspase 9
E. Pro-caspase 10

✔ Answer: E

The apoptosome (intrinsic pathway) = Cytochrome C + APAF-1 + dATP → forms a wheel-like structure that recruits and activates pro-caspase 9. Pro-caspase 10 is involved in the EXTRINSIC pathway (activated by DISC after death receptor ligation), not in the apoptosome. (★ Past paper: SA2 Med34 Q4, Key Simulation Q4, Re-exam Q8)

Q37. ★ [PAST PAPER] Which MAPK signalling pathway kinase induces apoptosis?

A. ERK1/2 (Extracellular signal-regulated kinase)
B. p38 MAPK
C. AKT (Protein kinase B)
D. PDK1
E. mTOR

✔ Answer: B

The MAPK family: ERK1/2 typically promotes cell survival and proliferation. p38 MAPK and JNK are stress-activated kinases that respond to UV, cytokines, osmotic stress — they promote apoptosis. AKT is a survival kinase (anti-apoptotic). (★ Past paper: SA2 Med34 Q3, Med36 Q5, Re-exam Q3)

Q38. ★ [PAST PAPER] Which of the following is a PRO-apoptotic protein?

A. Bcl-2
B. BAX
C. Mcl-1
D. Bcl-w
E. Bcl-xL

✔ Answer: B

Bcl-2 family: ANTI-apoptotic: Bcl-2, Bcl-xL, Mcl-1, Bcl-w (these have BH1-BH4 domains). PRO-apoptotic: BAX, BAK (multidomain), BAD, BID, PUMA, NOXA (BH3-only). BAX/BAK form pores in the mitochondrial outer membrane, releasing Cytochrome C. (★ Past paper: SA2 Med34 Q7, Key Simulation Q7, Re-exam Q2)

Q39. ★ [PAST PAPER] Which technique detects apoptosis by identifying nuclear morphology changes (chromatin condensation)?

A. Western blot
B. JC-1 staining (mitochondrial membrane potential)
C. Hoechst 33342 staining
D. Annexin V / PI flow cytometry
E. TUNEL assay

✔ Answer: C

Hoechst 33342 is a cell-permeable DNA dye that binds the minor groove of AT-rich DNA regions. Under fluorescence microscopy, apoptotic cells show condensed, fragmented, bright nuclei vs. normal diffuse nuclear staining. Flow cytometry with PI shows sub-G1 peak; Annexin V detects phosphatidylserine externalisation. (★ Past paper: SA2 Med34 Q5, SA2 MED35, Re-exam Q9)

Q40. ★ [PAST PAPER] In flow cytometric cell cycle analysis, apoptotic cells appear as which population?

A. G0/G1 peak
B. Sub-G1 peak
C. S phase fraction
D. G2/M peak
E. Hyperdiploid population

✔ Answer: B

During apoptosis, nuclear DNA is fragmented by endonucleases. These fragments can be washed out during staining, leaving cells with LESS DNA than normal G1 cells. When DNA content is measured by flow cytometry with PI, these cells appear as a 'sub-G1' peak to the left of the G1 peak. (★ Past paper: SA2 Med34 Q6, FA2 Sec11 slide, Re-exam Q10)

Q41. ★ [PAST PAPER] Which caspase is the KEY INITIATOR of the EXTRINSIC apoptosis pathway?

A. Caspase 3
B. Caspase 6
C. Caspase 7
D. Caspase 8
E. Caspase 9

✔ Answer: D

Extrinsic pathway: death receptor (Fas/TRAIL-R/TNF-R) ligation → DISC formation → recruitment and activation of pro-caspase 8 (initiator) → activated caspase 8 cleaves executioner caspases (3, 7) or BID (linking to intrinsic). Caspase 9 is the initiator of the INTRINSIC pathway. (★ Past paper: SA2 Med34, Key Simulation Q3)

Q42. Phosphatidylserine externalisation during apoptosis is detected by which reagent in flow cytometry?

A. Propidium iodide (PI)
B. Hoechst 33342
C. TUNEL reagent
D. Fluorochrome-labelled Annexin V
E. JC-1 dye

✔ Answer: D

In healthy cells, phosphatidylserine (PS) is on the inner leaflet of the plasma membrane. During EARLY apoptosis, PS flips to the outer surface (externalisation). Annexin V is a calcium-dependent phospholipid-binding protein with high affinity for PS. Fluorochrome-labelled Annexin V specifically binds exposed PS, marking early apoptotic cells.

LECTURE 15: LIVER FUNCTION

Q43. ★ [PAST PAPER] Which enzyme is the RATE-LIMITING step in bile acid synthesis?

A. HMG-CoA reductase
B. 7-alpha-hydroxylase (CYP7A1)
C. Sterol 27-hydroxylase
D. UDP-glucuronyl transferase
E. Heme oxygenase

✔ Answer: B

Cholesterol 7-alpha-hydroxylase (CYP7A1) catalyses the first and committed step in the classic pathway of bile acid synthesis. It is found ONLY in hepatocytes, is upregulated by cholesterol, and downregulated by bile salts (feedback inhibition). Statins target HMG-CoA reductase (cholesterol synthesis). (★ Past paper: SA2 MED35 Q66, SA2 Med34 Q4, Med36 Lec12 Q1)

Q44. ★ [PAST PAPER] A patient with jaundice and bile duct gallstones would most likely show elevation of which marker?

A. Serum albumin
B. Gamma-glutamyl transferase (GGT)
C. Alpha-fetoprotein (AFP)
D. Glucuronic acid
E. Aminotransferases (AST/ALT)

✔ Answer: B

Bile duct obstruction (post-hepatic/cholestatic jaundice) causes elevation of ALKALINE PHOSPHATASE (ALP) and GGT — enzymes from bile canaliculi epithelium that are released when bile backs up. ALT/AST are markers of hepatocellular INJURY. AFP is a liver cancer marker. (★ Past paper: SA2 MED35 Q63)

Q45. ★ [PAST PAPER] Which molecule conjugates with bilirubin to make it water-soluble?

A. Fatty acid
B. Acetyl CoA
C. Glucuronic acid
D. Sulphate
E. Albumin

✔ Answer: C

Unconjugated bilirubin is hydrophobic and transported in blood bound to albumin. In hepatocytes, UDP-glucuronyl transferase (UGT1A1) catalyses conjugation with glucuronic acid → bilirubin diglucuronide (conjugated, water-soluble) → excreted into bile. (★ Past paper: SA2 MED35 Q88, FA2 Med33-34 Q12, Re-exam Q3)

Q46. ★ [PAST PAPER] A liver cancer marker used clinically is:

A. Albumin
B. Prothrombin
C. Alpha-fetoprotein (AFP)
D. Haptoglobin
E. C-reactive protein

✔ Answer: C

AFP is normally produced by fetal liver but is very low in adults (<10 ng/mL). In adults, markedly elevated AFP suggests hepatocellular carcinoma (HCC), testicular cancer, or ovarian cancer. It is also elevated in pregnancy and some liver diseases (hepatitis, cirrhosis). (★ Past paper: SA2 Med34 Q6, Re-exam Q10)

Q47. ★ [PAST PAPER]   Disulfiram (Antabuse) inhibits which enzyme in ethanol metabolism?

A. Alcohol dehydrogenase
B. Aldehyde dehydrogenase (ALDH)
C. CYP2E1 (MEOS)
D. Acetyl-CoA synthetase
E. Pyruvate dehydrogenase

✔ Answer: B

Ethanol → (alcohol dehydrogenase) → Acetaldehyde → (aldehyde dehydrogenase, ALDH) → Acetate. Disulfiram blocks ALDH, causing acetaldehyde accumulation after alcohol consumption → flushing, tachycardia, nausea, vomiting (Asian flush-like reaction). This aversion is used therapeutically to discourage drinking. (★ Past paper: SA2 MED35, Med36, Re-exam Q19)

Q48. ★ [PAST PAPER]   Which phase of xenobiotic metabolism involves conjugation reactions (e.g., glucuronidation, glutathione conjugation)?

A. Phase I — Oxidation/Reduction/Hydrolysis
B. Phase II — Conjugation
C. Phase III — Excretion
D. Phase 0 — Absorption
E. Both Phase I and II simultaneously

✔ Answer: B

Phase I (CYP450-mediated): introduces a reactive group via oxidation, reduction, or hydrolysis. Phase II (transferases): conjugates the reactive group with glucuronic acid, glutathione, sulphate, acetyl-CoA, or amino acids → increases water-solubility for excretion. Phase II enzyme example: Glutathione S-Transferase (GST), UDP-glucuronyl transferase (UGT). (★ Past paper: SA2 Med34 Q3, Med36 Lec12 Q7)

Q49. ★ [PAST PAPER]   In thalassemia, the predominant form of elevated bilirubin is:

A. Conjugated bilirubin
B. Unconjugated bilirubin
C. Bilirubin bound to apolipoprotein
D. Bilirubin bound to bile acids
E. Bilirubin bound to globulin

✔ Answer: B

Thalassemia causes haemolysis (RBC destruction). Excess haemoglobin → heme → biliverdin → UNCONJUGATED bilirubin. This overwhelms the liver's conjugation capacity. PRE-hepatic (haemolytic) jaundice = elevated UNCONJUGATED bilirubin. (★ Past paper: FA2 Sec11 Q10, FA2 Med33-34 Q10, Re-exam Q1)

Q50. ★ [PAST PAPER]   Excess NADH generated by ethanol metabolism leads to which metabolic consequence?

A. Increased gluconeogenesis
B. Activated TCA cycle
C. Increased fatty acid synthesis (contributing to fatty liver)
D. Decreased lactic acid production
E. Increased beta-oxidation

✔ Answer: C

Ethanol → large amounts of NADH. High NADH/NAD+ ratio: (1) inhibits gluconeogenesis (OAA → malate); (2) inhibits TCA cycle (isocitrate DH, alpha-KG DH need NAD+); (3) inhibits fatty acid beta-oxidation; (4) promotes fatty acid SYNTHESIS (NADH drives reductive reactions); (5) promotes lactic acidosis. Net result: triglyceride accumulation → fatty liver → alcoholic hepatitis → cirrhosis. (★ Past paper: SA2 Med34 Q8, FA2 Sec11 Q11)

Q51. ★ [PAST PAPER] Which is a secondary bile acid?

A. Cholic acid
B. Chenodeoxycholic acid
C. Lithocholic acid
D. Glycocholic acid
E. Taurocholic acid

✔ Answer: C

PRIMARY bile acids (synthesised in liver): cholic acid and chenodeoxycholic acid. They are conjugated with glycine or taurine before secretion. SECONDARY bile acids (produced in intestine by bacterial deconjugation and dehydroxylation): deoxycholic acid (from cholic) and LITHOCHOLIC acid (from chenodeoxycholic). Glycocholic and taurocholic are conjugated forms of primary acids. (★ Past paper: SA2 Med34 Q7, FA2 Sec11 Q12)

Q52. ★ [PAST PAPER] Crigler-Najjar syndrome Type I results from deficiency of which enzyme?

A. Heme oxygenase
B. Biliverdin reductase
C. UDP-glucuronyl transferase (UGT1A1)
D. Cholesterol 7-alpha-hydroxylase
E. Alanine aminotransferase

✔ Answer: C

UGT1A1 normally conjugates bilirubin with glucuronic acid in hepatocytes. In Crigler-Najjar Type I, UGT1A1 is completely absent → severe unconjugated hyperbilirubinemia from birth → kernicterus (bilirubin deposits in brain) → fatal if untreated. Type II has reduced (but some) UGT1A1 activity → milder. Gilbert's syndrome = mild UGT1A1 reduction (10-30%) → usually asymptomatic. (★ Past paper: Med36 Lec12 Q6)

Q53. Which enzyme is responsible for alcohol metabolism via the Microsomal Ethanol Oxidising System (MEOS)?

A. Alcohol dehydrogenase (ADH)
B. Aldehyde dehydrogenase (ALDH)
C. CYP2E1
D. CYP3A4
E. Xanthine oxidase

✔ Answer: C

MEOS operates in the smooth ER (microsomes) and is induced at high ethanol concentrations or with chronic alcohol use. The key enzyme is CYP2E1 (a cytochrome P450). MEOS converts ethanol to acetaldehyde using NADPH and O2, generating free radicals (ROS), contributing to oxidative stress and liver damage.

Q54. Which of the following is the BEST indicator of liver SYNTHETIC function?

A. ALT (alanine aminotransferase)
B. GGT (gamma-glutamyl transferase)
C. Alkaline phosphatase (ALP)
D. Serum albumin and prothrombin time (PT)
E. Total bilirubin

✔ Answer: D

Liver synthetic function tests measure what the liver MAKES: (1) Albumin — major plasma protein, exclusively made by hepatocytes; (2) Prothrombin time (PT) — reflects clotting factors II, VII, IX, X made by liver. ALT/AST are hepatocellular INJURY markers. GGT/ALP are biliary markers. Bilirubin reflects excretory function.

LECTURE 16: BLOOD & HEME METABOLISM

Q55. ★ [PAST PAPER]   What is the RATE-LIMITING enzyme of heme synthesis?

A. ALA dehydratase (ALAD)
B. Delta-aminolevulinic acid synthase (ALAS)
C. Porphobilinogen deaminase
D. Ferrochelatase
E. Uroporphyrinogen decarboxylase

✔ Answer: B

ALAS (also called ALA synthase) catalyses the FIRST and rate-limiting step: glycine + succinyl-CoA → delta-aminolevulinic acid (ALA). It is regulated by heme (feedback inhibition) and requires pyridoxal phosphate (PLP, Vitamin B6). It occurs in the mitochondria. (★ Past paper: SA2 Med34 Q2, Re-exam Q6)

Q56. ★ [PAST PAPER]   Which TCA cycle compound is a PRECURSOR for heme synthesis?

A. Oxaloacetate
B. Citrate
C. Succinyl-CoA
D. Fumarate
E. Malate

✔ Answer: C

Heme synthesis begins with succinyl-CoA (from TCA cycle) + glycine → ALA (by ALAS in mitochondria). Succinyl-CoA is produced from alpha-ketoglutarate by alpha-KG dehydrogenase, from methylmalonyl-CoA, or from amino acid catabolism. (★ Past paper: FA2 Sec11 Q13, FA2 Med33-34 Q13, Re-exam Q11)

Q57. ★ [PAST PAPER]   Lead (Pb) poisoning inhibits which enzymes in heme synthesis?

A. ALAS and ferrochelatase
B. ALA dehydratase (ALAD) and ferrochelatase
C. Porphobilinogen deaminase and uroporphyrinogen decarboxylase
D. ALAS and uroporphyrinogen synthase
E. Ferrochelatase and porphobilinogen deaminase

✔ Answer: B

Lead inhibits TWO enzymes: (1) ALA dehydratase (ALAD) — normally converts 2 ALA → porphobilinogen; lead blocks this → ALA accumulates in urine (ALA is neurotoxic); (2) Ferrochelatase — normally inserts Fe2+ into protoporphyrin IX → heme; lead blocks this → zinc protoporphyrin accumulates. Result: microcytic anaemia with basophilic stippling. (★ Past paper: SA2 Med34 Q3, BCH212 Re-exam Q2)

Q58. ★ [PAST PAPER]   Porphyria Cutanea Tarda (PCT) is caused by deficiency of which enzyme?

A. ALAS
B. ALA dehydratase
C. Porphobilinogen deaminase
D. Uroporphyrinogen decarboxylase
E. Ferrochelatase

✔ Answer: D

PCT — the most common porphyria — is caused by deficient uroporphyrinogen decarboxylase (UROD). Uroporphyrinogens accumulate → photosensitivity (blistering, fragile skin), hyperpigmentation. NO neurological symptoms (unlike AIP). Triggered by alcohol, hepatitis C, iron overload, oestrogen. (★ Past paper: FA2 Sec11 Q14, FA2 Med33-34 Q13, SA2 Med34 Q, Re-exam Q1)

Q59. ★ [PAST PAPER] Acute Intermittent Porphyria (AIP) is characterised by all of the following EXCEPT:

A. Abdominal pain
B. Neuropsychiatric symptoms
C. Port-wine coloured urine
D. Photosensitivity
E. Precipitated by barbiturates

✔ Answer: D

AIP (porphobilinogen deaminase deficiency): ALA and PBG accumulate → NEUROLOGICAL symptoms (abdominal pain, neuropathy, psychiatric symptoms, autonomic dysfunction). NO photosensitivity — no porphyrins accumulate in the skin (unlike PCT). Port-wine urine from PBG oxidation. Barbiturates and other drugs induce ALAS, worsening attacks. (★ Past paper: Key Simulation Q, Re-exam Q8)

Q60. ★ [PAST PAPER] Which enzyme requires Vitamin B6 (pyridoxal phosphate) as a cofactor in heme synthesis?

A. ALA synthase (ALAS)
B. ALA dehydratase
C. Porphobilinogen deaminase
D. Uroporphyrinogen decarboxylase
E. Ferrochelatase

✔ Answer: A

ALAS catalyses: Glycine + Succinyl-CoA → ALA + CO2. This is a Schiff base condensation reaction requiring PLP (Pyridoxal Phosphate, derived from Vitamin B6) as cofactor. B6 deficiency can impair heme synthesis, contributing to sideroblastic anaemia. (★ Past paper: FA2 Sec11 Q15, BCH212 Re-exam Q4)

Q61. Which organ cannot synthesise heme?

A. Liver
B. Mature red blood cells (erythrocytes)
C. Heart muscle
D. Kidney
E. Bone marrow (erythroblasts)

✔ Answer: B

Mature red blood cells have no nucleus and no mitochondria. Since ALAS (first step of heme synthesis) is mitochondrial, and mature RBCs lack mitochondria, they CANNOT synthesise heme. Heme is synthesised during erythroblast development in the bone marrow, and the liver is the other major site. (★ Past paper: SA2 Med34 Q1, Re-exam Q22)

Q62. Which amino acid, together with succinyl-CoA, forms ALA in the first step of heme synthesis?

A. Alanine
B. Serine
C. Glycine
D. Tyrosine
E. Cysteine

✔ Answer: C

ALAS condenses glycine + succinyl-CoA → ALA (delta-aminolevulinic acid) + CO2. Glycine provides the carbon and nitrogen backbone. Succinyl-CoA provides 4 carbons. This occurs in the mitochondrial matrix.

LECTURE 17: PLASMA PROTEIN & BLOOD CLOTTING

Q63. ★ [PAST PAPER] What is the MAJOR plasma protein and its primary function?

A. Prealbumin — transports thyroxine
B. Albumin — maintains plasma oncotic (osmotic) pressure
C. Globulin — primary immune defence
D. Fibrinogen — blood clotting
E. Alpha-fetoprotein — fetal development

✔ Answer: B

Albumin constitutes ~60% of total plasma protein and has the highest concentration (~3.5–5 g/dL). Its primary function is to maintain colloid osmotic (oncotic) pressure, keeping fluid within blood vessels. It also transports fatty acids, bilirubin, hormones, and drugs. Synthesised exclusively by the liver. (★ Past paper: FA2 Sec11 Q18)

Q64. ★ [PAST PAPER] Which enzyme digests fibrin clots (fibrinolysis)?

A. Fibrinogen
B. Plasmin
C. Dicoumarol
D. Warfarin
E. Thrombin

✔ Answer: B

Plasmin is a serine protease that cleaves fibrin clots. Plasminogen (inactive) is converted to plasmin by tissue plasminogen activator (t-PA) and urokinase. This is fibrinolysis. Thrombin converts fibrinogen → fibrin (clot formation, opposite). Warfarin and dicoumarol inhibit vitamin K recycling. (★ Past paper: FA2 Sec11 Q17, Med36)

Q65. ★ [PAST PAPER] Which condition is associated with elevated serum gamma-globulin levels?

A. Liver cirrhosis (decreased)
B. Protein-losing enteropathy (decreased)
C. Immunodeficiency (decreased)
D. Multiple myeloma (markedly elevated)
E. Nephrotic syndrome (decreased)

✔ Answer: D

Gamma-globulins = immunoglobulins. Multiple myeloma is a plasma cell malignancy producing a monoclonal immunoglobulin (M-protein) → very high gamma-globulin (narrow 'M-spike' on electrophoresis). Conditions causing LOW gamma-globulin: immunodeficiency, protein-losing conditions (enteropathy, nephrotic syndrome), liver disease. (★ Past paper: Med36 Q1)

Q66. ★ [PAST PAPER] Alpha-1 antitrypsin (AAT) deficiency primarily causes which disease?

A. Multiple myeloma
B. Von Willebrand disease
C. Emphysema (and liver disease)
D. Haemophilia A
E. DIC

✔ Answer: C

AAT inhibits neutrophil elastase in the lungs. Without AAT, elastase degrades lung elastin → emphysema (panacinar type, especially lower lobes). Misfolded AAT also accumulates in hepatocytes → cirrhosis. AAT is an acute-phase protein made by the liver. (★ Past paper: Lec16-17 notes, Med36 Q3)

Q67. ★ [PAST PAPER] Aspirin prevents heart attacks by inhibiting which process?

A. Fibrinolysis
B. Prostaglandin and thromboxane synthesis (COX inhibition)
C. Vitamin K recycling
D. Thrombin formation
E. Antithrombin III

✔ Answer: B

Aspirin IRREVERSIBLY inhibits cyclooxygenase (COX-1 and COX-2) enzymes. This blocks synthesis of thromboxane A2 (TXA2) in platelets — TXA2 normally promotes platelet aggregation and vasoconstriction. Inhibiting TXA2 reduces platelet clumping → antiplatelet effect → prevents arterial thrombosis. (★ Past paper: SA2 MED35 Q11, Key Simulation Q2)

Q68. ★ [PAST PAPER] Which is the FIRST step in haemostasis after blood vessel injury?

A. Fibrin clot formation
B. Platelet aggregation
C. Blood vessel constriction (vasoconstriction)
D. Serotonin secretion from platelets
E. Clot degradation (fibrinolysis)

✔ Answer: C

Haemostasis sequence: (1) Vasoconstriction — reduces blood flow to injury site; (2) Primary haemostasis — platelet adhesion (via vWF-collagen-GP Ib/IX) and aggregation (GP IIb/IIIa-fibrinogen) → platelet plug; (3) Secondary haemostasis — coagulation cascade → fibrin reinforces plug; (4) Fibrinolysis — clot dissolution. (★ Past paper: FA2 Sec11 Q16, FA2 Med33-34 Q16)

Q69. ★ [PAST PAPER] Vitamin K deficiency increases which of the following?

A. Plasma concentration of prothrombin (Factor II)
B. Plasma concentration of free thrombin
C. Plasma concentration of calcitonin
D. Time for blood to clot (Prothrombin time, PT)
E. Time for broken bones to heal

✔ Answer: D

Vitamin K is required for gamma-carboxylation of glutamate residues in clotting factors II, VII, IX, X (and proteins C, S). Without this modification, factors cannot bind Ca^{2+} and are non-functional. Result: prolonged PT (tests extrinsic + common pathway). Warfarin works by blocking VitK recycling. (★ Past paper: SA2 MED35 Q45)

Q70. Von Willebrand factor (vWF) has which primary function in platelet adhesion?

A. Activates the intrinsic coagulation pathway
B. Bridges platelet GP Ib/IX receptor to exposed subendothelial collagen
C. Converts fibrinogen to fibrin
D. Inhibits thrombin
E. Activates protein C

✔ Answer: B

When a vessel is damaged, subendothelial collagen is exposed. vWF (multimeric glycoprotein from endothelium and platelets) binds to collagen AND simultaneously binds GP Ib/IX receptor on platelets — acts as molecular glue anchoring platelets to the injury site. vWF also carries and stabilises Factor VIII. Von Willebrand disease = most common inherited bleeding disorder.

LECTURE 18: NUTRITION & VITAMINS

Q71. ★ [PAST PAPER] Which vitamin deficiency causes Pellagra (dermatitis, diarrhea, dementia)?

A. Vitamin B1 (Thiamine)
B. Vitamin B2 (Riboflavin)
C. Vitamin B3 (Niacin)
D. Vitamin B6 (Pyridoxine)
E. Vitamin B12 (Cobalamin)

✔ Answer: C

Pellagra = Niacin (B3) deficiency. The 4 Ds: Dermatitis (photosensitive rash), Diarrhea, Dementia, Death. Niacin is needed for NAD+/NADH. It can be synthesised from tryptophan (60mg Trp → 1mg niacin), requiring B6. Isoniazid (TB treatment) inhibits B6 → secondary pellagra. (★ Past paper: SA2 Med34 Q1, BCH212 Re-exam Q36, Copy Q10)

Q72. ★ [PAST PAPER] Eating raw egg whites over a prolonged period causes deficiency of which vitamin?

A. Thiamine (B1)
B. Riboflavin (B2)
C. Niacin (B3)
D. Biotin (B7)
E. Folate (B9)

✔ Answer: D

Raw egg whites contain AVIDIN, a protein that tightly and irreversibly binds biotin (B7/H) in the gut, preventing its absorption. Cooking denatures avidin, allowing biotin absorption. Biotin is a cofactor for carboxylase enzymes (pyruvate carboxylase, acetyl-CoA carboxylase). Deficiency: alopecia, dermatitis, neurological symptoms. (★ Past paper: SA2 Med34 Nutrition Q, SA2 Med36 Q4, FA2 Med33-34 Q23, Re-exam Q5)

Q73. ★ [PAST PAPER] Macrocytic anaemia with elevated homocysteine but NORMAL methylmalonic acid indicates deficiency of:

A. Vitamin B12 (Cobalamin)
B. Folate (B9)
C. Vitamin B6
D. Vitamin C
E. Iron

✔ Answer: B

Both B12 and folate deficiency cause macrocytic anaemia and elevated homocysteine. KEY DISTINGUISHING TEST: methylmalonic acid (MMA). B12 deficiency → elevated MMA (because methylmalonyl-CoA mutase requires B12). Folate deficiency → NORMAL MMA. Folate deficiency also causes neural tube defects if during pregnancy. (★ Past paper: SA2 Med36 Q3, Re-exam Q52, Key Simulation Q1 Lec18)

Q74. ★ [PAST PAPER] Vitamin D deficiency causes which metabolic change?

A. Hypercalcaemia and hyperphosphataemia
B. Hypocalcaemia and hypophosphataemia (causing rickets/osteomalacia)
C. Neuromuscular irritability (only)
D. Night blindness
E. Megaloblastic anaemia

✔ Answer: B

Vitamin D → increases intestinal Ca^{2+} and phosphate absorption, kidney reabsorption, and bone mineralisation. Deficiency → low calcium (hypocalcaemia) and low phosphate (hypophosphataemia) → impaired bone mineralisation → Rickets (children, bowed legs) / Osteomalacia (adults, bone pain). Hypercalcaemia is seen with VitD TOXICITY. Neuromuscular irritability is a consequence of hypocalcaemia. (★ Past paper: SA2 Med34 Lec18 Q3)

Q75. ★ [PAST PAPER] Vitamin C is a cofactor for which enzyme in collagen synthesis?

A. Carboxylase
B. Prolyl/Lysyl Hydroxylase
C. Decarboxylase
D. Hydratase
E. Kinase

✔ Answer: B

Collagen synthesis requires hydroxylation of proline (→ hydroxyproline) and lysine (→ hydroxylysine) residues by prolyl hydroxylase and lysyl hydroxylase respectively. These enzymes require Vitamin C (ascorbic acid) as an electron donor to keep iron (Fe^{2+}) reduced (active). Deficiency → Scurvy: poor wound healing, bleeding gums, perifollicular haemorrhages. (★ Past paper: SA2 Med34 Q2, BCH212 Re-exam Q12, Copy Q6)

Q76. ★ [PAST PAPER] Which vitamin deficiency causes cretinism in fetal/newborn development?

A. Iron
B. Iodine
C. Folate
D. Vitamin B12
E. Zinc

✔ Answer: B

Iodine is essential for thyroid hormone (T3, T4) synthesis. Severe iodine deficiency during pregnancy → maternal and fetal hypothyroidism → cretinism: severe intellectual disability, stunted growth, deaf-mutism, spastic diplegia. Goitre (enlarged thyroid) results from iodine deficiency at any age. (★ Past paper: SA2 MED35 Q71, SA2 Med34 Lec21 Q7, Re-exam Q2)

Q77. ★ [PAST PAPER] Neural tube defects (spina bifida, anencephaly) are caused by deficiency of which vitamin?

A. Folate (B9)
B. Vitamin B12
C. Vitamin D
D. Vitamin A
E. Biotin

✔ Answer: A

Folate is critical for neural tube closure, which occurs 21-28 days after conception (often before a woman knows she is pregnant). Folate deficiency → failure of neural tube to close → spina bifida or anencephaly. ALL women of childbearing age should take 400-800 mcg folate/day. Encephalocele (another NTD) is also folate-related. (★ Past paper: SA2 MED35, Re-exam Q10, Copy Q7)

Q78. ★ [PAST PAPER] Isoniazid (INH) treatment for tuberculosis can lead to secondary deficiency of which vitamin?

A. Vitamin B1 (Thiamine)
B. Vitamin B3 (Niacin)
C. Vitamin B6 (Pyridoxine)
D. Vitamin B12
E. Vitamin C

✔ Answer: C

Isoniazid competes with pyridoxal phosphate (active B6) and inhibits pyridoxal kinase, preventing B6 activation. B6 deficiency leads to: peripheral neuropathy, secondary pellagra (because B6 is needed for niacin synthesis from tryptophan), sideroblastic anaemia. Pyridoxine (B6) supplements are given prophylactically with INH. (★ Past paper: BCH212 Re-exam Q15, Copy Q23, Lec18)

Q79. ★ [PAST PAPER] Which is the active form of Vitamin D in the body, produced in the kidney?

A. Cholecalciferol (Vitamin D3)
B. Ergocalciferol (Vitamin D2)
C. 25-hydroxycalciferol (calcidiol)
D. 1,25-dihydroxycalciferol (calcitriol)
E. 24,25-dihydroxycalciferol

✔ Answer: D

VitD pathway: Skin (UV) → Cholecalciferol (D3) → Liver (25-hydroxylase) → 25-OH-D3 (calcidiol, storage form, long half-life) → Kidney (1-alpha-hydroxylase) → 1,25-(OH)2-D3 (calcitriol, ACTIVE form, short half-life). Calcitriol acts like a steroid hormone. 25-OH-D3 is measured clinically (longer half-life). (★ Past paper: SA2 Med34 Lec21 Q9, BCH212 Re-exam Q6)

Q80. ★ [PAST PAPER] Kwashiorkor is characterised by edema and an enlarged liver with severe protein malnutrition. Which of the following signs is seen in Kwashiorkor but NOT in Marasmus?

A. Thin, wasted appearance
B. Oedema (swelling)
C. Loss of appetite
D. Low body weight
E. Dehydration

✔ Answer: B

Kwashiorkor = protein deficiency (often adequate calories): moon face, oedema (low albumin → low oncotic pressure → fluid leaks into tissues), enlarged fatty liver, skin lesions, apathy. Marasmus = total calorie deprivation: severe wasting/emaciation ('monkey face'), no oedema, child looks very thin. (★ Past paper: SA2 Med34 Lec21 Q8, Med36 Q4, Copy Q26, Re-exam Q3)

Q81. Prolonged use of proton pump inhibitors (PPIs) most likely leads to deficiency of which vitamin?

A. Vitamin B1
B. Folate
C. Vitamin B12
D. Vitamin D
E. Vitamin K

✔ Answer: C

Vitamin B12 requires gastric acid to separate it from food proteins, then binds intrinsic factor (IF) for ileal absorption. PPIs suppress gastric acid → B12 remains bound to food proteins and cannot bind IF → B12 malabsorption. Long-term PPI use (>2 years) significantly increases B12 deficiency risk. (★ Past paper: SA2 Med36 Q1, SA2 Med34 Lec21 Q4)

Q82. Which of the following about fat-soluble vitamins is TRUE?

A. They mostly act as coenzymes in metabolic pathways
B. They are stored in body fat and liver (can accumulate to toxic levels)
C. They are easily excreted in urine
D. They require frequent daily dietary intake
E. Excessive intake is safe

✔ Answer: B

Fat-soluble vitamins (A, D, E, K): stored in adipose tissue and liver, NOT excreted easily in urine. This means they can accumulate → hypervitaminosis if excessive intake (especially A and D). Water-soluble vitamins (B complex, C) act as coenzymes, are easily excreted in urine, and require regular intake. (★ Past paper: FA2 Sec11 Q19)

LECTURE 19: GENETIC ENGINEERING

Q83. ★ [PAST PAPER]   Which technique is MOST appropriate for detecting SARS-CoV-2 viral LOAD (quantitative measurement)?

A. ATK (Antigen Test Kit)
B. PCR (conventional)
C. RT-PCR
D. RT-qPCR (reverse transcription quantitative PCR)
E. ELISA

✔ Answer: D

SARS-CoV-2 has an RNA genome. RT-qPCR: (1) Reverse transcriptase converts viral RNA → cDNA; (2) qPCR quantifies DNA in REAL TIME using fluorescent dye/probe. The Ct (cycle threshold) value is inversely proportional to viral load. RT-PCR can detect but not QUANTIFY. ATK detects antigen (qualitative). ELISA detects antibodies. (★ Past paper: FA2 Sec11 Q22, Med36 Q1, SA2 MED35 mRNA question)

Q84. ★ [PAST PAPER]   In a forensic crime scene, DNA obtained from suspects and victims is analysed using VNTR probes. What technique is used?

A. Northern blot
B. Southern blot
C. Western blot
D. RT-PCR
E. DNA microarray

✔ Answer: B

Southern blot: DNA is digested with restriction enzymes, separated by gel electrophoresis, transferred to a membrane, then probed with VNTR (Variable Number Tandem Repeat) probes. This creates a DNA 'fingerprint' unique to each individual (used in forensics and paternity testing). Modern methods use PCR-based STR analysis (CODIS). (★ Past paper: SA2 MED35 Q29, SA2 Med34 Q2 Lec19, Re-exam Q27)

Q85. ★ [PAST PAPER]   Which technique is used for high-throughput gene expression pattern analysis in cancer research?

A. FISH (Fluorescence In Situ Hybridisation)
B. RT-PCR
C. PCR-RFLP
D. DNA microarray
E. Sanger sequencing

✔ Answer: D

DNA microarray simultaneously measures expression of thousands of genes. mRNA is extracted, reverse transcribed, labelled with fluorescent dye, and hybridised to the array. Spots that light up indicate expressed genes. Used to identify gene expression profiles in cancer vs normal tissue. (★ Past paper: FA2 Sec11 Q23)

Q86. ★ [PAST PAPER] What method introduces foreign DNA into cells using an electrical field?

A. Electrophoresis
B. Electroporation
C. Heat shock transformation
D. Chromatography
E. Centrifugation

✔ Answer: B

Electroporation applies a brief high-voltage electric pulse to cells. This creates temporary pores in the cell membrane, allowing charged DNA to enter (DNA moves toward anode). Pores reseal after pulse stops. Used for cells that are difficult to transform by other methods. Heat shock transformation uses Ca2+ + rapid heating to allow DNA uptake. (★ Past paper: SA2 Med36 Q4)

Q87. ★ [PAST PAPER] Why is RT-PCR used to clone a EUKARYOTIC gene for expression in BACTERIA?

A. Bacteria grow faster than eukaryotes
B. Bacteria can splice out introns from eukaryotic genes
C. RT-PCR generates intron-free cDNA from mRNA, which bacteria can express
D. PCR cannot work with eukaryotic DNA
E. Bacteria require RNA as template

✔ Answer: C

Eukaryotic genes contain introns (non-coding sequences). Bacteria LACK the spliceosome machinery to remove introns. RT-PCR: mRNA (already spliced, intron-free) → reverse transcriptase → cDNA → PCR amplification → cDNA without introns → can be expressed in bacteria. This produces the correct protein product. (★ Past paper: Genetic Engineering lecture, Copy Q39)

Q88. ★ [PAST PAPER] DNA ligase creates which type of bond when joining two DNA fragments?

A. Hydrogen bond
B. Phosphodiester bond
C. Peptide bond
D. Glycosidic bond
E. Van der Waals interaction

✔ Answer: B

DNA ligase seals the sugar-phosphate backbone by forming a phosphodiester bond between the 3'-OH of one nucleotide and the 5'-phosphate of the adjacent nucleotide. This requires NAD+ (E. coli ligase) or ATP (T4 ligase) as energy source. T4 ligase joins both sticky ends and blunt ends; E. coli ligase joins only sticky ends. (★ Past paper: Key Simulation Lec19 Q2, Re-exam Q30)

Q89. ★ [PAST PAPER] Which is the FASTEST method to screen bacterial colonies for successful gene insertion?

A. Replica plating
B. Colony PCR
C. RT-PCR
D. Southern blotting
E. Sanger sequencing

✔ Answer: B

Colony PCR: directly pick cells from a colony → add to PCR mixture → run PCR with insert-specific primers → check gel. A positive band = correct insert present. This takes hours vs. days for DNA extraction-based methods. It's the fastest screening method. (★ Past paper: Key Simulation Lec19 Q4, Re-exam Q32)

Q90. ★ [PAST PAPER] Western blotting uses which reagent to detect specific protein fragments?

A. Fluorescent DNA probe
B. VNTR probe
C. Antibody specific to the target protein
D. Complementary hybridised nucleic acid probe
E. Restriction enzyme

✔ Answer: C

Western blot (immunoblotting): proteins are separated by SDS-PAGE, transferred to a PVDF membrane, blocked, then incubated with PRIMARY antibody (specific to target protein) → secondary antibody (enzyme-conjugated) → substrate → produces detectable signal. Used to detect protein size and expression levels. Northern blot uses RNA probes; Southern blot uses DNA probes. (★ Past paper: Key Simulation Lec19 Q5, Re-exam Q33)

Q91. A restriction enzyme cuts DNA at the sequence 5'-GAATTC-3'. What type of ends does it produce?

A. Blunt ends
B. 5' sticky ends (overhangs)
C. 3' sticky ends (overhangs)
D. Nick in one strand only
E. Circular ends

✔ Answer: B

EcoRI cuts 5'-G|AATTC-3' / 3'-CTTAA|G-5', producing 5' overhangs (sticky ends): 5'-AATT-3'. These single-stranded overhangs can hybridise with complementary sticky ends from another fragment cut with the same enzyme, facilitating ligation. Blunt-end cutters: SmaI, EcoRV, HincII.

Q92. Huntington's disease involves which type of mutation in the HTT gene?

A. Missense point mutation
B. Large deletion of exons
C. Excessive CAG trinucleotide repeat expansion
D. Promoter methylation causing gene silencing
E. Nonsense mutation creating premature stop codon

✔ Answer: C

Huntington's disease is caused by expansion of CAG repeats (>36 repeats) in exon 1 of the HTT gene (chromosome 4). CAG codes for glutamine, so a polyglutamine (polyQ) tract is formed in the huntingtin protein. This toxic gain-of-function protein accumulates and causes progressive neurodegeneration. CRISPR/Cas9 NHEJ can be used to silence the mutant allele. (★ Past paper: SA2 MED35 on Huntington's, Genetic Engineering lecture)

LECTURE 20: FREE RADICALS & ANTIOXIDANTS

Q93. ★ [PAST PAPER] Which enzyme provides the FIRST LINE of defence against superoxide radicals?

A. Catalase
B. Glutathione peroxidase
C. Superoxide dismutase (SOD)
D. Glutathione reductase
E. Peroxiredoxin

✔ Answer: C

SOD catalyses: $2O_2^{\bullet-} + 2H^+ \rightarrow H_2O_2 + O_2$. This converts toxic superoxide to less reactive H_2O_2. Three isoforms: Cu/Zn-SOD (cytoplasm), Mn-SOD (mitochondria), extracellular SOD. The H_2O_2 is then handled by catalase ($2H_2O_2 \rightarrow 2H_2O + O_2$) or glutathione peroxidase. SOD = first enzymatic antioxidant defence. (★ Past paper: SA2 Med34 Lec20 Q1, FA2 Sec11 Q27, Re-exam Q3)

Q94. ★ [PAST PAPER] Which mineral is incorporated into glutathione peroxidase (GPx)?

A. Iron (Fe)
B. Manganese (Mn)
C. Selenium (Se)
D. Copper (Cu)
E. Magnesium (Mg)

✔ Answer: C

GPx is a selenium-dependent enzyme containing selenocysteine (instead of cysteine) at its active site. GPx reduces H_2O_2 and lipid hydroperoxides (ROOH) using glutathione (GSH) as electron donor: $2GSH + H_2O_2 \rightarrow GSSG + 2H_2O$. Selenium deficiency → increased oxidative damage. Manganese is in Mn-SOD; Iron in catalase; Copper/Zinc in Cu/Zn-SOD. (★ Past paper: FA2 Sec11 Q26, FA2 Med33-34 Q20, Re-exam Q2)

Q95. ★ [PAST PAPER] Which vitamin regenerates oxidised Vitamin E back to its active form?

A. Vitamin A
B. Vitamin C (Ascorbic acid)
C. Vitamin D
D. Vitamin K
E. Vitamin B12

✔ Answer: B

Vitamin E (alpha-tocopherol) donates an electron to neutralise a lipid peroxyl radical → tocopheroxyl radical (inactive). Vitamin C (water-soluble) then donates an electron to regenerate active Vitamin E. This vitamin C-E regeneration cycle is important in protecting membranes. Glutathione then regenerates Vitamin C. (★ Past paper: SA2 Med34 Lec20 Q3, Re-exam Q4)

Q96. ★ [PAST PAPER] Which of the following is a characteristic property of free radicals EXCEPT:

A. Extreme reactivity
B. Small, diffusible molecules
C. Short lifespan
D. Long lifespan (hours-days)
E. Generate new radicals by chain reaction

✔ Answer: D

Free radicals have an unpaired electron making them highly reactive (A), small and diffusible (B), with extremely SHORT lifespans (fractions of a second to microseconds) (NOT long) (C), and they propagate chain reactions by stealing electrons from neighbouring molecules (E). Option D is incorrect — their lifespan is very short, not long. (★ Past paper: SA2 Med34 Lec20 Q2, Re-exam Q5)

Q97. ★ [PAST PAPER] Which part of amino acids is PRIMARILY attacked by free radicals?

A. R-group (side chain)
B. OH group
C. Amino group (-NH2)
D. Carboxylic group (-COOH)
E. Sulfhydryl group (-SH) of cysteine

✔ Answer: E

Sulfhydryl groups (-SH) in cysteine residues are particularly susceptible to oxidation by ROS, forming disulfide bonds (S-S), mixed disulfides with glutathione (RS-SG), or sulphenic/sulphinic acid. This can alter protein structure and function. Aromatic residues (Tyr, Trp, Phe) are also oxidised. (★ Past paper: FA2 Sec11 Q25, FA2 Med33-34 Q19, Re-exam Q1)

Q98. Which antioxidants are classified as CELL MEMBRANE antioxidants?

A. Vitamin C and uric acid
B. SOD and catalase
C. Alpha-tocopherol (Vitamin E) and Coenzyme Q10
D. Transferrin and ceruloplasmin
E. Glutathione peroxidase and glutathione reductase

✔ Answer: C

Classification by location: PLASMA antioxidants: Vitamin C, uric acid, bilirubin, beta-carotene, albumin, ceruloplasmin, transferrin. CELL MEMBRANE antioxidants: alpha-tocopherol (Vitamin E) and Coenzyme Q10 — both are fat-soluble, embedded in membranes, protecting phospholipids from lipid peroxidation. INTRACELLULAR antioxidants: SOD, catalase, GPx.

LECTURE 21: CLINICAL CORRELATION (BIOCHEMICAL & DISEASES)

Q99. ★ [PAST PAPER] Which of the following oils has the HIGHEST proportion of SATURATED fatty acids?

A. Corn oil
B. Coconut oil
C. Olive oil
D. Canola oil
E. Soybean oil

✔ Answer: B

Coconut oil is ~90% saturated fatty acids (mainly lauric, myristic, caprylic — medium-chain). Olive oil is ~75% monounsaturated (oleic acid). Canola, soybean, corn are rich in polyunsaturated fatty acids. High saturated fat → raises LDL cholesterol → cardiovascular risk. (★ Past paper: SA2 Med34 Lec21 Q1, Re-exam Q1, BCH212 Re-exam)

Q100. ★ [PAST PAPER] What is the ANTIDOTE for paracetamol (acetaminophen) overdose?

A. EDTA
B. Deferoxamine (DFO)
C. Naloxone
D. Fomepizole
E. N-acetylcysteine (NAC)

✔ Answer: E

Paracetamol overdose: CYP2E1 produces toxic NAPQI (N-acetyl-p-benzoquinone imine) which depletes hepatic glutathione → hepatocyte necrosis. NAC replenishes glutathione (substrate for synthesis) and directly inactivates NAPQI. Must give within 8-10 hours for maximum benefit. Naloxone: opioid antidote. Fomepizole: methanol/ethylene glycol antidote. (★ Past paper: SA2 Med36 Q2, SA2 Med34 Lec21 Q5, Re-exam Q8)

Q101. ★ [PAST PAPER] Which of the following is secreted in bile AND urine after phototherapy for neonatal jaundice?

A. Urobilinogen
B. Stercobilinogen
C. Bilirubin
D. Lumirubin
E. Urobilin

✔ Answer: D

Phototherapy uses blue light to convert unconjugated bilirubin (water-insoluble) into LUMIRUBIN, a structural isomer that is water-soluble WITHOUT needing hepatic conjugation. Lumirubin can be excreted directly in both bile and urine. This is why neonatal jaundice improves with phototherapy even with immature conjugation enzymes. (★ Past paper: SA2 Med36 Q1, SA2 Med34 Lec21 Q3, Re-exam Q9)

Q102. ★ [PAST PAPER]   Band keratopathy of the cornea is associated with which condition?

A. Obesity
B. Diabetes mellitus
C. Hypocalcaemia
D. Hypercalcaemia
E. Hypothyroidism

✔ Answer: D

Band keratopathy = calcium-phosphate deposits in the corneal epithelium and Bowman's layer, appearing as a white/grey band across the cornea. It is associated with HYPERCALCAEMIA (e.g., hyperparathyroidism, Vitamin D toxicity, sarcoidosis, chronic kidney disease with high phosphate). Can also occur with chronic ocular inflammation. (★ Past paper: SA2 Med36 Q3, Lec21 clinical case)

Q103. ★ [PAST PAPER]   Which of the following endocrine disrupting chemicals (EDCs) can be found in plastics?

A. Dioxins
B. Lead
C. Bisphenol A (BPA)
D. Mercury
E. Arsenic

✔ Answer: C

Bisphenol A (BPA) is used in polycarbonate plastics and epoxy resin linings. It leaches from containers and acts as a xenoestrogen (mimics oestrogen) — an endocrine disruptor. BPA is also found in phthalates (plasticisers in PVC). Lead, mercury, arsenic are heavy metal EDCs; dioxins are from industrial/combustion processes. (★ Past paper: SA2 Med34 Lec21 Q6, Re-exam Q4)

Q104. ★ [PAST PAPER]   Which form of Vitamin D has the LONGEST half-life in the bloodstream?

A. 1,25-dihydroxycalciferol (calcitriol)
B. 25-hydroxycalciferol (calcidiol)
C. Vitamin D2 (ergocalciferol)
D. Vitamin D3 (cholecalciferol)
E. 7-dehydrocholesterol

✔ Answer: B

25-hydroxyvitamin D (25-OH-D, calcidiol) has a half-life of 2-3 weeks — it is the STORAGE form and the form measured clinically to assess vitamin D status. The active form 1,25-(OH)2-D (calcitriol) has a very short half-life (4-6 hours). D3 and D2 dietary forms are rapidly converted. (★ Past paper: SA2 Med34 Lec21 Q9, Copy Q82, BCH212 Re-exam)

Q105. ★ [PAST PAPER]   Bitot's spots on the conjunctiva are an EARLY sign of which vitamin deficiency?

A. Vitamin D deficiency
B. Vitamin A deficiency
C. Vitamin C deficiency
D. Vitamin K deficiency
E. Vitamin B12 deficiency

✔ Answer: B

Vitamin A deficiency eye signs (progressive): Night blindness (earliest) → Conjunctival xerosis (dryness) → Bitot's spots (white, foamy triangular patches of keratinised cells on conjunctiva) → Corneal xerosis → Corneal ulceration/keratomalacia → Blindness (irreversible). (★ Past paper: Lec21 clinical, SA2 MED35 Q35, Copy Q61, Re-exam Q20)

Q106. ★ [PAST PAPER]   A 50-year-old man with alcoholic liver disease presents with vision problems at night, dry eyes, and skin lesions. What is the most likely deficiency?

A. Vitamin B1 (Thiamine)
B. Vitamin B3 (Niacin)
C. Vitamin A (Retinol)
D. Vitamin C (Ascorbic acid)
E. Vitamin K

✔ Answer: C

Alcoholic liver disease impairs synthesis of retinol-binding protein (RBP) and reduces Vitamin A storage in the liver. Night blindness (11-cis retinal needed for rhodopsin), xerophthalmia, and skin changes result from Vitamin A deficiency. Poor dietary intake in alcoholics compounds the problem. (★ Past paper: Lec21 clinical case 1)

Q107. ★ [PAST PAPER]   A patient with a history of 10 years of alcohol use presents with leg oedema, difficulty breathing, and cardiomegaly. What is the most likely diagnosis?

A. Pellagra
B. Wet beriberi (Thiamine deficiency)
C. Vitamin B12 deficiency
D. Vitamin D deficiency
E. Scurvy

✔ Answer: B

WET BERIBERI = Thiamine (B1) deficiency with CARDIOVASCULAR features: high-output cardiac failure, oedema, cardiomegaly, tachycardia. Alcoholics are at high risk due to poor diet and impaired absorption. Thiamine is needed for pyruvate dehydrogenase → pyruvate accumulates → lactic acidosis + impaired ATP synthesis in heart. DRY beriberi = peripheral neuropathy. (★ Past paper: Lec21 clinical case 3)

Q108. ★ [PAST PAPER]   Pernicious anaemia is caused by:

A. Decreased dietary iron intake
B. Autoimmune destruction of gastric parietal cells → lack of intrinsic factor → B12 malabsorption
C. Folate deficiency due to poor diet
D. Haemolysis of red blood cells
E. Renal failure causing decreased erythropoietin

✔ Answer: B

Pernicious anaemia: autoantibodies against gastric parietal cells and/or intrinsic factor (IF) → parietal cells destroyed → no HCl, no IF → B12 cannot be absorbed from terminal ileum → macrocytic anaemia + subacute combined degeneration of the spinal cord (neurological damage). Treatment: lifelong parenteral (injection) B12. (★ Past paper: Lec21 clinical case 6&7, Lec18)

Q109. ★ [PAST PAPER] A 5-year-old child presents with bowed legs and easy fractures. Calcium is low, phosphate is low, PTH is elevated. What is the deficient nutrient?

A. Calcium directly
B. Vitamin D (Calciferol)
C. Phosphate directly
D. Vitamin C
E. Vitamin K

✔ Answer: B

Rickets in children: Vitamin D deficiency → decreased intestinal Ca2+ and phosphate absorption → hypocalcaemia + hypophosphataemia → secondary hyperparathyroidism (PTH rises to raise Ca2+) → impaired bone mineralisation → bowed legs, frontal bossing, rachitic rosary, delayed fontanelle closure. (★ Past paper: SA2 Med34 Lec18 Q4, Lec21 clinical case 7)

Q110. ★ [PAST PAPER] An elderly woman on prolonged antibiotics develops bruising and elevated PT/INR. What is the most likely cause?

A. Vitamin C deficiency
B. Platelet dysfunction
C. Vitamin K deficiency (due to reduced gut bacteria)
D. Haemophilia A
E. Von Willebrand disease

✔ Answer: C

Prolonged antibiotics kill intestinal bacteria that synthesise Vitamin K2. Combined with poor diet → Vitamin K deficiency → impaired gamma-carboxylation of factors II, VII, IX, X → prolonged PT/INR → bleeding tendency. Newborns are also at risk (sterile gut + poor VitK in breast milk). (★ Past paper: Lec21 clinical case 8, Lec18)

Q111. ★ [PAST PAPER] In phenylketonuria (PKU), which amino acid is decreased in plasma?

A. Alanine
B. Cysteine
C. Tyrosine
D. Glutamine
E. Phenylalanine

✔ Answer: C

PKU: phenylalanine hydroxylase (PAH) deficiency → cannot convert Phe → Tyrosine. Phenylalanine ACCUMULATES (hyperphenylalaninaemia → neurotoxic). Tyrosine becomes CONDITIONALLY ESSENTIAL (cannot make it from Phe) → DECREASED. Tyrosine is the precursor for dopamine, epinephrine, thyroxine, and melanin. (★ Past paper: SA2 MED35 Q16, Lec15 Human Genome)

Q112. An OGTT (Oral Glucose Tolerance Test) uses what dose of glucose?

A. 25 g
B. 50 g
C. 75 g
D. 100 g
E. 150 g

✔ Answer: C

Standard OGTT: patient fasts 8 hours, baseline blood glucose measured, then drinks 75g glucose in 250-300mL water. Blood glucose measured at 1 and 2 hours. Diagnostic criteria for diabetes: 2-hour glucose ≥200 mg/dL. For gestational diabetes screening, 50g or 100g are sometimes used. (★ Past paper: SA2 MED35 Q80)

Q113. A patient has Kayser-Fleischer rings in the cornea, liver disease, and neuropsychiatric symptoms. What is the diagnosis and which protein is decreased?

A. Haemochromatosis — transferrin decreased
B. Wilson's disease — ceruloplasmin decreased
C. Alpha-1 antitrypsin deficiency — AAT decreased
D. Familial hypercholesterolaemia — LDL receptor decreased
E. Porphyria cutanea tarda — uroporphyrinogen decarboxylase decreased

✔ Answer: B

Wilson's disease (autosomal recessive, ATP7B gene mutation): impaired copper excretion into bile → copper accumulates in liver, brain, cornea, kidney. KF rings = copper deposits in Descemet's membrane of cornea. Ceruloplasmin (copper-binding protein) is markedly LOW. Treatment: penicillamine (copper chelator) or zinc (blocks copper absorption). (★ Past paper: Lec21 clinical case 17)

Q114. A 40-year-old man with chronic gout develops tophaceous deposits in joints and soft tissues. The primary biochemical abnormality is:

A. Elevated uric acid (hyperuricaemia) from purine metabolism
B. Elevated calcium in blood
C. Elevated triglycerides
D. Decreased uric acid excretion from calcium pyrophosphate deposition
E. Elevated homocysteine

✔ Answer: A

Gout: hyperuricaemia → monosodium urate crystal deposition in joints (especially 1st MTP), bursae, and soft tissues (tophi). Uric acid is the end product of purine catabolism (xanthine oxidase). Causes: overproduction (high purine diet, lesch-nyhan) or underexcretion (renal disease, diuretics, alcohol). Pseudogout = calcium pyrophosphate crystals. (★ Past paper: Lec21 clinical case 18)

Q115. A female patient from an iodine-deficient area presents with goitre, heat intolerance, palpitations, and weight loss. What is the most likely diagnosis?

A. Hypothyroidism from iodine deficiency
B. Hyperthyroidism (toxic nodular goitre) from prolonged iodine deficiency causing autonomy
C. Hashimoto's thyroiditis
D. Vitamin A deficiency
E. Pernicious anaemia

✔ Answer: B

Initially, iodine deficiency → hypothyroidism → TSH rises → goitre. Chronically, nodules in the goitre can become autonomously functioning (independent of TSH) → hyperthyroidism (heat intolerance, palpitations, weight loss, sweating). Iodine sufficiency prevents both conditions. Graves' disease is autoimmune hyperthyroidism. (★ Past paper: Lec21 clinical case 19)

Q116. A 14-year-old obese child has acanthosis nigricans (dark velvety patches) on the neck and axillae. This finding is most closely associated with:

A. Vitamin A toxicity
B. Insulin resistance and hyperinsulinaemia
C. Thyroid hormone excess
D. Cortisol deficiency
E. Growth hormone excess

✔ Answer: B

Acanthosis nigricans (AN) = hyperpigmented, velvety, hyperkeratotic skin in body folds (neck, axillae, groin). In young obese individuals, BENIGN AN is caused by insulin resistance → hyperinsulinaemia → insulin binds IGF-1 receptors on keratinocytes → epidermal proliferation. Associated with metabolic syndrome, type 2 diabetes. MALIGNANT AN in older patients: sudden onset → suspect visceral malignancy (especially gastric cancer). (★ Past paper: Lec21 clinical case 12)

END OF SIMULATION EXAM

Summary of Questions

Lecture	Topic	Questions	Past Paper Qs
Lec 10	DNA & RNA Synthesis	8	7
Lec 11	Protein Synthesis & Transport	8	7
Lec 12	Human Genome & Gene Regulation	10	8
Lec 13	Thalassemia	8	7
Lec 14	Apoptosis	8	7
Lec 15	Liver Function	12	10
Lec 16	Blood & Heme Metabolism	8	7
Lec 17	Plasma Protein & Blood Clotting	8	6
Lec 18	Nutrition & Vitamins	12	10
Lec 19	Genetic Engineering	10	8
Lec 20	Free Radicals & Antioxidants	6	5
Lec 21	Clinical Correlation	18	12
TOTAL		**116***	**94**

** Note: This exam contains 116 questions. ★ marks indicate questions drawn from past papers — study these with extra attention!*